

S 02

UNIT - 01 Cognitive and Concept Development in Children

* **Cognitive Development:** Cognitive development refers to the process of acquiring sensory information, thinking about it, and gradually making children capable of using it in various situations to solve different kinds of problems.

Cognitive development is used interchangeably with intellectual development. Intellectual development is a continuous process. It includes patterned responses, which involve mental behaviors such as sensation, perception, memory, reasoning, and imagination. In the intellectual development of children, sensation and perception have special significance.

According to **Reber**:- "Cognition refers to those mental behaviors whose nature is abstract and which include representation, subtle expectation, complex rule utilization, imagery, belief, intention, problem-solving, and others."

"Developing all those mental abilities and capacities of the child, as a result of which he can adjust to his changing environment and dedicate/apply himself to the solution of difficult and serious problems."

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Characteristics of Cognitive Development:

The characteristics of cognitive development are as follows:

1. Cognitive development refers to intellectual development.
2. Cognitive development is an individual process that moves from general to specific.
3. Cognition is an acquired ability that continues throughout life and also brings about adjustment with the environment.
4. For cognitive development, the development and maturity of the sensory organs are essential.
5. Through this, the learning capacity in humans increases.
6. It is through cognition that a human sees, understands, experiences, and acquires related knowledge.
7. Its development begins even before infancy.

Steps in the Cognitive Development Process:

It consists of six steps:-

1. Attention: Attention is the process of focusing on an object. This object can be in both concrete and abstract forms.

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2. Perception/Belief: Perception is a process of acquiring information from the surrounding environment and processing it. The child builds perceptions towards concepts by acquiring information through his sensory organs and senses (eyes, ears, nose, mouth, and fingers).

3. Memory: Memory is the process of storing, searching, and recalling knowledge. The child stores learned knowledge in his memory and, upon the presentation of an appropriate stimulus, recalls the corresponding knowledge and responds.

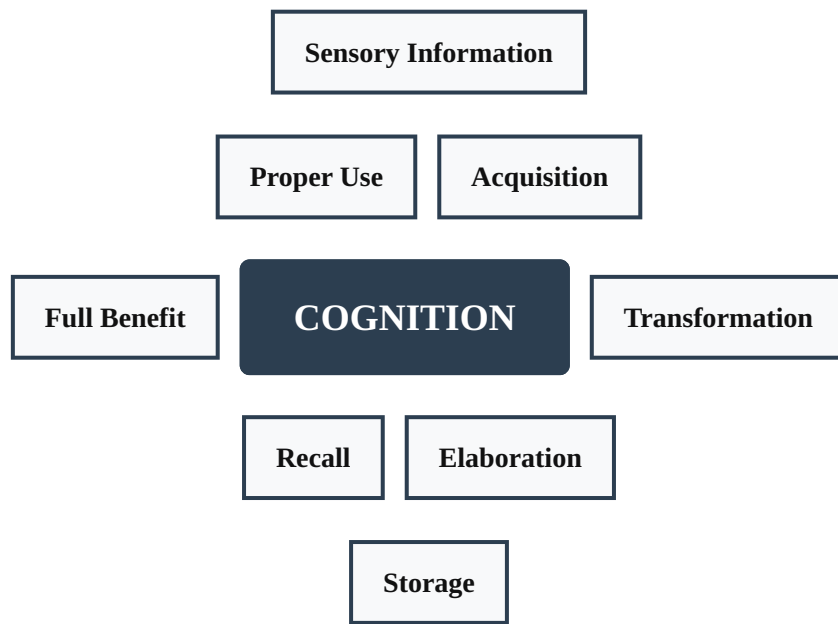
4. Language: Language is the process of developing understanding through reading, writing, speaking, listening, and dialogue. Language has various dimensions, such as stability, heuristic capacity, practical use, and it is helpful in the development of necessary cognitive processes.

5. Learning: Learning is a synthetic process of curiosity towards new knowledge and acquiring it.

6. Higher Reasoning: Reasoning is a process of reflective cognition. It includes problem-solving planning, logic, decision-making, etc.

* **Cognition:** Cognition is a process in which mental operations such as sensation, perception, imagery, retention, problem-solving, reasoning, thinking, and recall are inherent.

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(Proper Use, Full Benefit, Sensory Information, Acquisition, Transformation, Elaboration, Cognition, Recall, Storage)

Role of Cognition in Education:-

The role of cognition in education is as follows -

1. Only through cognition can an individual express his feelings before everyone.
2. Cognition also plays a crucial role in acquiring education and in teaching.
3. Through cognition alone do we obtain knowledge of our surrounding environment.
4. Cognition has an important contribution to the acquisition of education from primary to higher levels.
5. Through cognition alone do we achieve intellectual development. Teaching is not possible without intellectual development.

Socio-Cultural Factors Affecting Cognition and Learning:

The socio-cultural factors affecting cognition and learning are as follows:-

- 1. Family:** A peaceful and pleasant environment of the family accelerates the mental development of the child. If the child's family is happy, educated, and civilized, then the child's mental ability will also develop in a smooth manner.
- 2. Socio-Economic Status:** Due to economic conditions, a child is sometimes unable to acquire cognition in a proper way. A rich child acquires more cognition compared to a poor child. The reason for this is the availability of resources provided to the child.

4. Health: If the child's health is not good, it directly impacts the child's mental development and learning process.

5. Education of Parents: If parents are educated, the cognitive development of the child will take place in an excellent manner.

6. School: If the environment of the school is appropriate and the teaching methodology is student-centered, it will have a good impact on the cognitive development of children.

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7. Society: The place where the child lives. If his environment is good and the surrounding people are civilized, the child's cognitive development will take place in a positive manner.

8. Teacher-Student Relationship: Having a good relationship between the child and the teacher enables the child to learn better and easily.

* Jean Piaget's Theory of Cognitive Development -:

Piaget was born on August 9, 1896, in Switzerland. Piaget is considered the pioneer of developmental psychology. Piaget studied cognitive development on an extensive scale.

"According to Piaget, the nature of the store of knowledge acquired by a child changes and gets redefined at each stage of development."

Piaget applied his cognitive theory of child development to his own children. Piaget is called the father of constructivism and the father of child psychology.

Process and Structure of Cognitive Development:-

1. Scheme:- (A Single Information) A single behavior towards any person or object.

2. Schemes:- An organized pattern of behaviors that can be easily repeated.

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"It is the expressed form of thinking carried out by a child for problem-solving."

3. Schema: A mental structure that can be generalized. (An organized form of information).

Piaget considered two main factors important in the process of cognitive development:-

(I) Organization

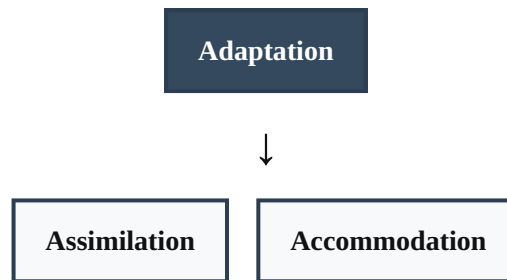
(II) Adaptation

(I) Organization: Organization implies that perception and intellectual information are arranged in such a way so that adjustment with the external environment can be achieved.

(II) Adaptation: The innate tendency to create harmony with the environment. "Adaptation is the process in which a child tries to adjust to his external environment. This is an innate tendency."

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1. Assimilation: This is the process of integrating a new object or event into existing experiences. For example, when a child is given a toffee, he puts it in his mouth because he knows that this object is meant to be eaten.

2. Accommodation: Accommodation is the process in which a child tries to bring modification in new experiences based on past experiences.

* **Equilibrium / Equilibration** -: This is the process through which a balance is established between assimilation and accommodation.

* **Cognitive Structure:** A collection/set or group of mental abilities is called a cognitive structure.

* **Conservation:** The ability to recognize and understand both change and stability in the environment, and the ability to differentiate it from changes in the shape, color, or form of an object.

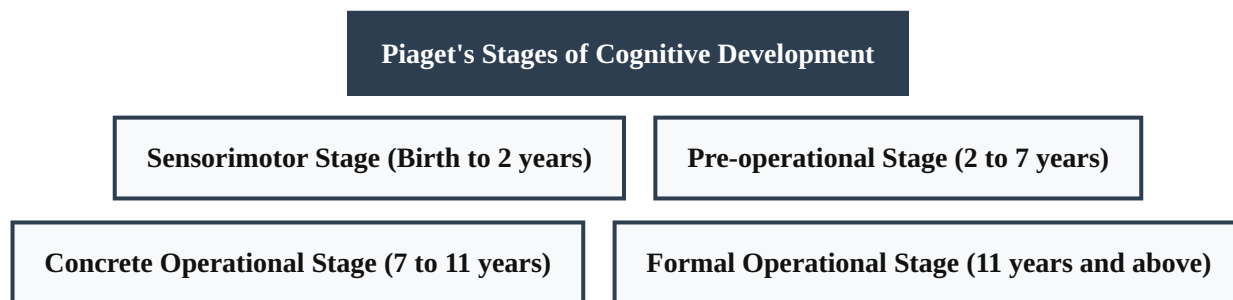
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* **Classification:** The ability to categorize on the basis of any characteristic such as color, shape, and size.

* **Seriation/Gradation:** The capacity to arrange objects in order according to any of their attributes, such as length, weight, etc.

Stages of Cognitive Development: Piaget described four stages of cognitive development.



1. Sensorimotor Stage -

- (I) This is the first stage of cognitive development.
- (II) This stage lasts from birth to 2 years.
- (III) In this stage, the child performs only simple actions.

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- (iv) In this stage, the child experiences objects, sounds, etc., with the help of his sensory organs.
- (v) In children of this stage, understanding develops through sensory experiences and by performing motor actions upon them.

2. Pre-operational Stage-

- (I) This is the second stage of cognitive development.
- (II) This stage lasts from 2 years to 7 years.
- (III) In this stage, the child begins to imagine.
- (iv) In this stage, the child learns through play and imitation.
- (v) In this stage, the child begins to acquire language.
- (vi) Children also start to think.
- (vii) Children learn by rote / memorization.
- (viii) In this stage, the child is self-centered.
- (ix) Gradually, he learns to select symbols.
- (x) Children consider their thoughts to be correct; they think that the entire world revolves around them. Piaget termed this egocentrism.

3. Concrete Operational Stage:-

- (i) This is the third stage of cognitive development.
- (ii) This stage ranges from 7 to 11 years.
- (iii) In this stage, the development of the capacity for logical reasoning begins.
- (iv) At this age, the child starts to think logically.

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- (v) Knowledge of the states of matter is acquired, i.e., understanding of solids, liquids, weight, etc.
- (vi) In this stage, the child learns to carry out conservation, relation, and classification.

4. Formal / Abstract Operational Stage:-

- (i) This is the fourth and final stage of cognitive development.
- (ii) Its duration is considered to be from 11 years and above.
- (iii) In this stage, the child possesses the capacity to understand discrepancies.
- (iv) The child begins to form hypotheses of events and also tries to verify them.
- (v) At this age, the child carries out problem-solving.
- (vi) In this stage, the child's brain begins to mature.

Characteristics of Piaget's Theory:-

The characteristics of Piaget's theory of cognitive development are as follows:

- (i) According to Piaget, a learning environment and activity are required for learning to take place.
- (ii) Learning is a gradual and ascending process.
- (iii) The power to think and discover in children is determined by their biological maturation, experience, and the interaction between these two.

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- (iv) Abstract thinking of children is the effect of their education. If a child's education is of a low standard, abstract thinking is also of a low standard, and if it is high, the children's thinking is also high.

(v) After the formal operational stage, the child's entire intellectual capacity is developed, and he becomes capable of solving problems in a systematic and logical manner.

Educational Utility of Piaget's Theory -:

- (i) In Piaget's theory, the learner, i.e., the child, remains in an active role. While preparing the curriculum, teachers should pay special attention to the needs and interests of the learner.
- (ii) Teachers should try to remove the defects arising in children's thinking, such as animism and egocentrism.
- (iii) According to Piaget, cognitive development in conservation, relation, classification, etc., takes place in children during the concrete operational stage. Teachers should conduct activities to enhance the development of this intellectual level so that creative, inventive, and critical qualities can develop in children.
- (iv) Imitation and play have educational value; we should present ideals so that children can imitate them, and cognitive development can take place through various games.

* Intelligence -:

First of all, in 1885, a French psychologist used the word intelligence. The literal meaning of intelligence is—different types of abilities. Such as:- learning and understanding quickly, good memory, logical thinking through which life's problems can be solved.

In 1905, Alfred Binet first worked to measure intelligence. In 1923, Boring gave a formal definition of intelligence that, "Intelligence is what intelligence tests measure." Intelligence is such a common term that we use more in our day-to-day conversational language, and we use the word intelligence for qualities like learning, understanding, and good memory power.

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- According to **Woodworth**:- "Intelligence is a method of acting."
- According to **Terman**:- "Intelligence is the ability to think in terms of abstract ideas."
- According to **Ross**:- "Conscious adaptation to a new situation is intelligence."
- According to **Thorndike**:- "The ability to profit from past experiences is intelligence."

Nature of Intelligence:-

1. Ability to Learn: According to Dearborn, "Intelligence is the capacity to learn or to profit by experience." Indian sages and scholars have considered knowledge as the main end and means of life. Therefore, society gives a high place to the person who acquires maximum knowledge.

2. Ability to Solve Problems: According to Rayburn, "Intelligence is the power which gives us the capacity to solve problems and achieve our objectives."

3. Ability to Think Abstractly: According to Terman, "An individual is intelligent in proportion as he is able to carry on abstract thinking."

4. Ability to Adjust to the Environment: According to William Stern, "Intelligence is a general mental adaptability to new problems and conditions of life."

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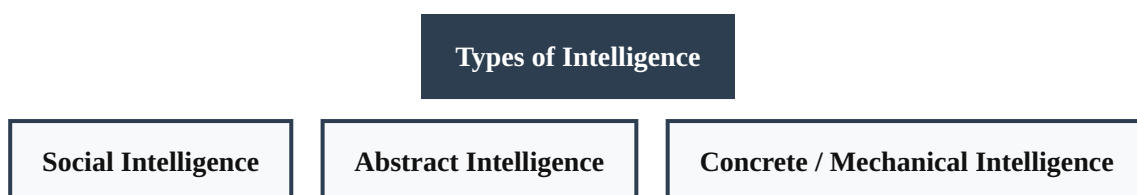
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Characteristics of Intelligence:-

- (i) Intelligence is an innate power.
- (ii) For the proper development of intelligence, environment is important.
- (iii) Through yogic practices, increase in innate intelligence is possible.
- (iv) Intelligence is the ability to learn.
- (v) It is the ability to think abstractly.
- (vi) Intelligence is the ability to profit from experiences and previously acquired knowledge.
- (vii) There is a difference between intelligence and knowledge.
- (viii) Intelligence is the ability to solve problems.
- (ix) It is influenced by heredity and environment.
- (x) The development of intelligence takes place from birth until the middle of adolescence.
- (xi) It is the ability to adjust to the environment.
- (xii) Intelligence is a developing and changing ability.

Types of Intelligence:-

Garrett and Thorndike have described three types of intelligence:-



(1) Social Intelligence:- This intelligence is related to social adaptability. According to Drever, "Social intelligence is a type of intelligence which is inherent in a person's behavior towards other individuals and social relationships."

(2) Abstract Intelligence: This intelligence is related to bookish knowledge. This kind of capacity is found more in philosophers, artists, and storytellers. Individuals who have more abstract intelligence become successful artists, painters, mathematicians, and storytellers.

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(3) Concrete Intelligence: This intelligence is related to tools and machines; it is called motor or mechanical intelligence. Garrett termed it concrete intelligence. Children who have more concrete intelligence possess a greater capacity for handicrafts. Concrete intelligence is found in engineers or skilled craftsmen.

Development of Intelligence:-

Development is a continuous and comprehensive process. After birth, physical development and mental development take place in accordance with the various stages of growth. Just as the physical development of a child is completed by the end of adolescence, similarly the mental development of the child, i.e., intellectual development, also achieves completion by the end of adolescence. Therefore, according to the stages of development, the description of the development of a child's intelligence is as follows:-

(1) Development of intelligence in Infancy: In this stage, the child perceives through his senses during the first two years of birth. Through these experiences alone, the child's mental development takes place. The development of intelligence takes place at a rapid pace between 3 to 5 years. Therefore, the child asks various types of questions to gain knowledge. In this stage, the tendency to imitate is higher in the child. He learns by imitating the behavior of his elders.

(2) Development of intelligence in Childhood: The most important stage of intellectual development is the child's childhood stage. In this stage, the child's intelligence develops at a rapid pace. About 75% of the child's intelligence develops in this very stage.

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(3) Development of intelligence in Adolescence: In this stage, the development of intelligence moves at a slow pace towards maturity. In this stage, all the mental powers of the adolescent are developed. In this stage, they acquire full capability for learning, attention, abstract thinking, adjustment, and problem-solving.

Determinants of Intelligence:

The determinants of intelligence are as follows:-

(1) Heredity: Heredity is the most prominent factor among the determinants of intelligence because heredity has the maximum impact on intelligence.

(i) Even after being brought up in an identical environment, genetically different children show differences in their intelligence quotient (IQ).

(ii) Similarity is visible in the intelligence of children with the intelligence of their parents.

(iii) A high degree of similarity is found in the intelligence of identical twins.

(iv) Not much similarity is found in the intelligence of fraternal twins.

(2) Environment: Environment also impacts the development of intelligence.

(i) Children who have been brought up in a rich and good environment have a better intelligence quotient compared to those children who have been brought up in poverty.

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(ii) When normal siblings were kept in an identical environment, the similarity in their intelligence was found to be higher compared to those normal siblings who were kept in different environments.

(3) Gender: Gender also plays an important role in affecting a child's intelligence. If girls do not get a proper environment, their thinking skills do not develop well. On the other hand, if boys get a proper environment, their thinking skills develop well.

(4) Health: If physical health is good, mental development is also good. If physical health is not good, the development of the child's intelligence will also not take place in a proper way.

(5) Age: Age also plays an important role in determining a child's intelligence. The development of a child's intelligence takes place only up to the age of 16 to 20 years.

Measurement of Intelligence (Intelligence Quotient):

The intelligence quotient (IQ) was first used by Terman.

$$\text{Intelligence Quotient (IQ)} = (\text{Mental Age} / \text{Chronological Age}) \times 100$$

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Values and Meaning of Intelligence Quotient (IQ):

IQ Range	Classification
Above 140	Gifted / Genius
120 - 139	Very Superior / Sharp
110 - 119	Superior / High Average
90 - 109	Normal / Average
80 - 89	Dull / Backward
70 - 79	Borderline / Feeble-minded
60 - 69	Moron / Moronity
20 - 59	Imbecile
Below 25	Idiot

Theories of Intelligence:-

1) Single-Factor / Uni-Factor Theory - Its pioneer is Alfred Binet, and Stern and Terman supported it. It consists of one element. It is also called the autocratic theory.

2) Two-Factor Theory - Spearman propounded this theory in 1904 AD. In 1927, he described two elements of intelligence:

i) General Factor (G-Factor)

ii) Specific Factor (S-Factor)

1) General Factor (G-Factor)

I. This factor is innate in humans.

II. This factor remains common in all actions.

III. Every individual differs in the general factor.

IV. Individuals who have more G-factor are more successful in life compared to others.

2) Specific Factor (S-Factor)

I. This factor is found in a lesser or greater amount in different individuals.

II. An individual who has more S-factor is proficient in that particular area. For example—art, dance, etc.

III. The specific factor is learned.

IV. This factor also generates individual differences.

3) Multi-Factor Theory - In 1926, American psychologist Thorndike refuted Spearman's two-factor theory related to intelligence and presented the multi-factor theory. According to Thorndike, intelligence is composed of numerous independent elements.

According to him, as many mental actions as a human performs, that many elements come under intelligence.

He compared his theory to a 'heap of sand'. On this basis, it is called the 'quantity theory'. Thorndike also described some core elements in which he utilized:

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I. Directional ability

II. Logical ability

III. Lecture/Verbal ability

IV. Numerical/Score ability

V. Remembering power and verbal ability.

4) Group-Factor Theory - This was propounded by Thurstone in 1938. He described 13 elements, out of which 7 elements were described as primary:

I. Verbal factor (V)

II. Number factor (N)

III. Word fluency factor (W)

IV. Memory factor (M)

V. Reasoning factor (R)

VI. Spatial factor (S)

VII. Perceptual factor (P)

$$\text{Intelligence} = V + N + W + M + R + S + P$$

5) Gardner's Theory of Multiple Intelligences :- In 1983, neuro-psychologist Gardner clarified on the basis of research that the form of intelligence is not single but multi-factorial.

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Gardner originally described seven types of intelligence. He added the eighth type in 1998 and the ninth type in 2000.

According to Gardner, at present there are the following nine types of intelligence -

- i) Verbal or Linguistic Intelligence
- ii) Spatial Intelligence
- iii) Musical Intelligence
- iv) Logical-Mathematical Intelligence
- v) Bodily-Kinesthetic Intelligence
- vi) Intrapersonal (Personal Self) Intelligence
- vii) Interpersonal (Personal Other) Intelligence
- viii) Naturalistic Intelligence
- ix) Existentialistic Intelligence

Gardner stated that several types of intelligences are found in most normal individuals, but sometimes due to heredity or environment, a particular intelligence becomes more developed, but its impact does not fall on other types of intelligence.

*** Concept Development (Samprayatay Vikas)**

A concept is an idea that relates generally and universally to a particular class, object, or group. For example—by saying 'human', it does not imply a single man or individual, but rather refers to the entire human race.

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- According to **Woodworth** - Concepts are thoughts that refer to objects, events, qualities, etc.
- According to **Boring, Langfeld, and Weld**:- "A concept is a general idea that expresses a general class."

The characteristics of concepts are as follows –

- i) It is the basis of the learning process.
- ii) It is a process of making learning simple. It is a general idea that expresses a general class.
- iii) It provides meaning to ideas obtained through imagination or perception.
- iv) Concepts depend on individual differences, which depend on a human's capacity to think, analyze, and generalize.
- v) In the beginning, concepts are vague but become clear through the experience of knowledge.

Types of Concepts -

1) Conjunctive Concept - Conjunctive concepts are rich in various characteristics. This type of concept settles easily and quickly in the brain. For example—cat, dog, cow.

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According to Deese:- "All those concepts that settle quickly in the mind are called conjunctive concepts."

2) Disjunctive Concept – Those concepts whose values and characteristics are used in place of one another. For example—two triangles, two rectangles, etc. This type of concept, even if equal in number, can change in its shape and form.

3) Relational Concept - Under this come those concepts between whose characteristics and values there is a definite relationship.

According to Bruner—an image enclosed within a boundary line, two images enclosed within boundary lines, and similarly a third image enclosed within a third boundary line. In this way, we can say that a relationship is established between the number of boundary lines and the number of images.

Development of Concepts in Children:

Children describe only what is in front of them, what they see, i.e., concrete concepts develop in children. With increasing age, they begin to know or tell about things that are not present in front of them, i.e., abstract concepts begin to develop or get developed.

The concepts that develop in children are as follows:-

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ii) **Spatial** – From two years itself, a child begins to have knowledge of this. With increasing age, this knowledge becomes permanent and clear. A child of three or four years begins to identify long and short objects.

"Concepts of distance, direction, height, length, depth, shape, etc., are called spatial concepts."

iii) **Shape** – At the stage of six months, the identification of shapes begins in children, and by the stage of two years, the child begins to differentiate between various shapes.

iv) **Time** – Knowledge of time develops in a child at the stage of 6 to 8 years. When a child starts going to school, he begins to have knowledge of time. In the beginning, they identify morning, afternoon, evening, and night.

v) **Number** – Knowledge of numbers begins to develop in children after the age of 5 years. When a child begins to go to school, it develops at a rapid pace through the teaching of mathematics.

Factors Affecting Concepts:-

i) Intellectual capacity, ii) Sensory organs, iii) Opportunities for learning, iv) Gender, v) Experience, vi) Maturity, vii) Thinking, viii) Information processing, ix) Socio-economic level.

Bruner's Theory of Cognitive Development:

Bruner gave a model of cognitive development. According to him, this is the model through which a human establishes harmony with his environment.

Bruner first conducted his study related to cognition on adults, after that on school-going children, then on three-year-old children, and then on newborn infants.

Bruner wrote a book named 'The Process of Education' in 1960 AD. This book was written on education, problems of education, and the presentation of ideas.

According to Bruner - "Learning is a goal-centered action that satisfies the curiosity of the learner."

Stages of Intellectual Development :-

On the basis of Piaget's cognitive development, Bruner also described the stages of intellectual development. He described 3 stages of intellectual development:-

1) Enactive Stage (Birth to 3 years): Performing muscular motor activities actively and interacting with the environment. In this stage, the child learns only through direct experience and by performing actions himself.

2) Iconic Stage (3-7 years): This stage lasts from 3 to 7 years. In this stage, the child's intellectual capacity becomes so developed that the child begins to think and understand in a concrete form.

3) Symbolic Stage (7 years and above): This stage extends from 7 years and above. Bruner termed this stage an important stage of intellectual development. At this age, children express their experiences through language.

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Bruner's Theory involves -

1) Analytical thinking versus Intuitive thinking, 2) Readiness, 3) Utility of self-action by learners, 4) Heuristic/Discovery learning, 5) Importance of connectedness, 6) Structure of the lesson.

The contribution of Bruner's theory to education is as follows:-

I. Language development, II. Discovery method, III. Genetic learning, IV. Teaching method, V. New role of the teacher, VI. Concept, VII. Intellectual development, VIII. Explanation of learning principles.

UNIT-2 :- Child Development and Learning

Growth and Development:-

> **Growth** - The word growth is the Hindi translation of the English word 'Growth', which means moving towards maturity. Growth implies quantitative changes in length, weight, shape, and various parts of the human body. Its scope is limited. Growth is merely a step of development. In growth, changes take place in both size and result, and its measurement is possible.

According to **Frank**:- "Growth refers to the increase taking place in cells."

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> **Development** - The word development is the Hindi translation of the English word 'Development'. In development, change is not purely quantitative. Development takes place mentally. We cannot measure or weigh development. Development is a lifelong process that continues from conception throughout life. The process of development includes physical, mental, social, emotional, and other aspects.

According to **Ira. J. Gordon**:- "Development is a process that goes on from a person's birth until that time when he achieves full development."

Characteristics of Development-

- i) Development is influenced by both environment and heredity.
- ii) Development is a progressive process.
- iii) Development is a maturity-oriented process.
- iv) Development is a continuous and gradual process.
- v) As a result of development, new characteristics and abilities manifest in an individual.
- vi) At the time of a child's birth, development takes place at a rapid pace and slows down in adulthood.

Factors Affecting Growth and Development:-

- i) Effect of heredity/inheritance, ii) Environmental effect, iii) Social effect, iv) Mental effect, v) Emotional effect, vi) Effect on character development, vii) Biological factor, viii) Physical factor, ix) Intelligence, x) Gender, xi) Nutrition, xii) Race, xiii) Economic status.

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Educational Significance of the Theory of Growth and Development :-

- i) From the theory of growth and development, we get to know that the pace and quantity of growth and development are not identical in all children.
- ii) For a child's growth and development, heredity and environment are equally responsible. Development is not possible without either one.
- iii) Various aspects of growth and development, such as mental development, emotional development, social development, etc., are interconnected. We get to focus on the holistic development of children.
- iv) Estimating the future progress of growth and development in children provides us the benefit that we should make efforts keeping in mind what the child can do or what he can become.

Interrelationship between Child Development and Learning:-

Learning has a direct relationship with development. A human begins to learn from birth itself, and it continues throughout life. As age increases, a human changes his behaviors along with his experiences. Learning impacts the physical and mental development of a human. A human definitely learns something or other at each stage of his development, but the pace of learning is not identical in each stage.

At the primary level, a child develops and tries to learn something or other. Some behaviors come to children hereditarily, and children learn some basic activities on their own, such as walking, running, sitting, eating-drinking, etc.

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Development requires a safe environment for learning. Children need to express themselves. Providing a safe environment to children is the responsibility of the school, home, and community.

Learning and Maturation :-

Learning - The process of learning is continuous and goes on throughout life. The more a person learns, the more rapidly his development takes place. The process of learning depends on continuity and universality.

- According to **Crow and Crow**: "The acquisition of habits, knowledge, and attitudes is also considered learning."
- According to **Skinner**: "Learning is a process of progressive behavior adaptation."
- According to **Guilford**: "Learning is any change in behavior resulting from behavior."

Nature and Characteristics of Learning:

i) Learning continues throughout life, ii) Learning is change, iii) Learning is universal, iv) Learning is adjustment, v) Learning is development, vi) Learning is the organization of experiences, vii) Learning is purposeful, viii) Learning is an active process, ix) Learning is discovery, x) Learning is a product of the environment.

Methods of Learning -

i) Learning by doing, ii) Learning by imitation, iii) Learning by observation, iv) Learning by testing/experimentation, v) Learning through collective/group methods.

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Factors Affecting Learning:

i) Physical and mental health, ii) Maturation, iii) Desire to learn, iv) Motivation, v) Environment, vi) Method of learning, vii) Role of the teacher, viii) Social factors, ix) Geographical factors, x) Psychological factors.

Maturation - Maturation is a process that comprehensively affects the stages of growth and development in a child. There is a deep relationship between maturation and learning. These are two sides of the same coin because almost every action of learning is based purely on maturation.

Maturation is of two types - 1. Physical, 2. Mental.

Both mental and physical maturation are essential for children. When a child begins to write, the muscles of his hands are trained so that the hand can work easily in the formation of words.

- According to **Bigge and Hunt**:- "Maturation is a developmental process under which an individual with time acquires all those characteristics and traits whose foundation was already laid in his cells at the time of conception."

Characteristics of Maturation:

- i) Maturation is related to physical changes over time.
- ii) Quantitative and qualitative changes take place in maturation.
- iii) Maturation is a process of biological and functional development of the cells of any organism.

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- iv) Maturation is a mental, physical, and natural process that is necessary before the appearance of any acquired behavior or before the acquisition of any behavior.

Difference between Learning and Maturation -:

Learning

- i) Learning is related to the actions of various organs of the body.
- ii) Learning is an acquired action.
- iii) Through learning, individual characteristics progress.
- iv) Practice has great importance in learning.
- v) Through the process of learning, changes are brought about in environmental conditions.

Maturation

- i) Maturation is not related to their actions, but to the structure of those organs.
- ii) Through maturation, chemical changes take place in physical structure.
- iii) Through maturation, racial characteristics advance, such as hair growth, teething, muscle development, etc.
- iv) Practice is not required in maturation.
- v) Maturation is an innate/natural action.

* Learning Disabilities and Abilities:

Learning ability implies how quickly a child learns a thing, and how well and easily he understands the lesson taught by the teacher.

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Learning disability implies that these children cannot be identified as quickly as exceptional children are identified. Learning disabled children live a normal life and do not face any kind of difficulty in their daily chores. They face difficulties only related to thinking, understanding, and language.

The identification of learning disabled children can be done in three ways:-

- 1) Lower educational achievement compared to normal intelligent children and the absence of any specificity in seeing or hearing.
- 2) Through the difference between ability and achievement.
- 3) Different types of performance related to various areas. "Children require a healthy body and a healthy mind to learn. A child requires physical and mental powers; children in whom these powers are less than average are called children with low learning capacity or slow learners."

According to **Hammill, Leigh, and McNutt**:- "Learning disability is such a mechanism where an individual experiencing significant difficulties in the acquisition and use of listening, speaking, reading, writing, reasoning, or mathematical abilities. These difficulties are intrinsic to the individual."

Learning Disabilities:

1. Dyslexia - Disability in reading.
2. Dysgraphia - Problem related to writing.
3. Dyscalculia - Problem related to calculation.

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4. Dyspraxia - Problem related to motor skills.
5. Dysmorphia - Considering oneself more beautiful, tall, or powerful.
6. Dysthymia - Problem of serious stress.
7. Aphasia - Problem related to language communication.

8. Progeria - Looking old at a young age.
9. Dementia - Inability to reason, weak memory power.
10. Bulimia - Disability related to eating habits/tendencies.
11. ADHD - Disability related to attention (attention deficit).

Assessment of Learning and Assessment for Learning:-

Assessment of Learning

- 1) In the assessment of learning, results are given through numerical grades, such as A, B, C.
- 2) In the assessment of learning, a student's achievement is compared against standards.
- 3) The result of the assessment of learning is communicated to parents or guardians, students, etc.
- 4) The assessment of learning takes place at the end of the completion of a unit.

Assessment for Learning

- 1) Assessment for learning consists of two steps: Diagnostic and Formative Assessment.

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- 2) No grades or marks are given in this. This paves the way for teachers to guide students to move forward in the right direction and for proper guidance and adjustment for them.
- 3) Assessment for learning can be based on various types of sources such as portfolios, work progress sheets, teacher observation, and interaction.

UNIT-3 Understanding of Behavioristic and Information Processing Theories of Learning

Pavlov's Theory of Classical Conditioning:

The theory of classical conditioning was propounded by the Russian scientist Pavlov in 1904.

In his experiment, Pavlov made a dog hungry and tied it to a table. He operated on the salivary glands of that dog. To collect saliva from the dog's mouth, it was connected to a glass jar through a tube. After that, Pavlov gave food to the dog. Seeing the food, saliva came into the dog's mouth, which is a natural process. This experiment was repeated several times, and the collected saliva was measured.

Experiment:- In the second step, Pavlov gave food along with ringing a bell (artificial stimulus) to the dog. Seeing the food, saliva came into the dog's mouth again, which is a natural response. Pavlov repeated this experiment also multiple times.

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In the final step, Pavlov only rang the bell. Hearing the sound of the bell, saliva came into the dog's mouth. The dog had learned that when the bell rings, food is received. Due to this very effect of learning, saliva started appearing repeatedly from his mouth upon ringing the bell.

From this, it concludes that when an action is repeated multiple times, the same natural response occurs even with an unnatural stimulus or artificial stimulus!

This experiment of Pavlov is called the conditioned response theory or classical conditioning theory.

Skinner's Theory of Operant Conditioning / Active Conditioning:-

This theory was propounded by American psychologist Prof. B. F. Skinner in 1938. Skinner conducted experiments on white rats and pigeons.

For the experiment on the rat, a box was created whose name was the Skinner Box. In this box, the target had to be reached through a dark, narrow path. Skinner enclosed a hungry rat in this box. Due to hunger, the rat jumped here and there. The arrangement in that box was such that when the rat was on the appropriate path, and as soon as the rat's foot hit a lever, a clicking sound would occur, light would appear, and then the rat would receive food. The rat repeated this action of pressing the lever, light appearing, and receiving food multiple times. The rat learned the art/action of pressing the lever to obtain food and light. Pressing the lever, the occurrence of the clicking sound or the appearance of light, and getting food worked as reinforcement.

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Skinner conducted the second experiment on a pigeon. He enclosed a hungry pigeon in a box. The pigeon sat quietly for a while. As soon as light appeared in the box, the pigeon started pecking, due to which it received a grain of corn. The pigeon performs this action repeatedly and learns to peck by turning its head. In this process, operant active conditioning develops.

On the basis of this theory of Skinner, emphasis is laid on dividing the subject matter into small lessons, making learning rapid and effective.

This theory states that if students are made aware of the results of their efforts, students can make more progress in their work. Reinforcement has immense importance in active conditioning. Reinforcement can take many forms, such as punishment, reward, knowledge of results, etc.

Information Processing Model:- Information processing implies the method or theory of learning. This is a cognitive theory that examines how information enters memory, is stored, and is retrieved from it. This model emphasizes memory.

According to this, three stores are found in memory, and received information flows after being processed in them.

1) Sensory Memory: Information enters here first, and the storage of information is for a very short duration, after which the information disappears.

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2) Working Memory: This is called short-term memory. In this stage as well, information cannot stay for a long time, at most 15-20 seconds. Under this, an individual responds as soon as he receives information.

3) Long-Term Memory: Information stored in this can remain accumulated for an indefinite period. Therefore, this memory is called a relatively permanent memory, and this memory has the maximum utility in human life.

UNIT – 4 Role of Society in Children's Development and Learning:

Interrelationship between Learning and Society:-

Learning and society are complementary to each other. Both are required for an individual's development. Development is not possible without either one. A child first learns from his parents or other members of the family. Thereafter, the child imitates the behaviors of his elders. He does exactly what he sees. After the family, he goes into society. There also he imitates, he watches the behaviors of the people there. Society desires that every individual of society become a skilled, well-adjusted, and good citizen. There is an intimate relationship between learning and society.

- According to **MacIver**:- "Society is a web of social relationships and it is always changing."

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- According to **LaPiere**: "Society refers not to a group of individuals, but to the complex system of interactions taking place among the individuals of the group."

Social Learning Theory (Bandura):

Albert Bandura propounded the theory of the Bobo Doll in 1961. This experiment was conducted on a large doll named Bobo Doll. It was carried out among children of 3-6 years.

In his social learning theory (1977), Bandura, contrary to Skinner, considered humans as active information processors. An individual seeks a relationship between behavior and its result. He considers that very behavior of the model (child) correct which he adopts. Bandura described four mediation processes/steps in his theory:

- 1) Attention - Imitation to focus attention
- 2) Retention - Specific memory process
- 3) Reproduction - Connecting with one's capacity or memory from time to time
- 4) Motivation - Willpower to perform the behavior.

Bandura believes that children observe various ways of behavior of the people around them. The child performs exactly that behavior which society considers appropriate for him.

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For Bandura's Bobo doll experiment, three types of groups were formed:

The first group was given an aggressive model.

The second group was given a non-aggressive model.

The third group was not given any model.

According to Bandura - "An individual observes the behaviors of others and integrates them into his own behavior to repeat them in the future."

Experiment:- In this experiment, children were shown a five-minute film. Three groups were formed for this. In the film, a large room had a lot of toys kept inside it. Out of those, one toy was a very large doll (Bobo Doll). An adult individual entered that room and started showing his aggression towards that large doll (Bobo Doll). He hits the doll, throws it on the floor, and kicks it.

Thereafter, this film was prepared in three forms:

In one film, the first group of children saw that the child (model) performing aggressive behavior was rewarded.

In the second film, the second group of children saw that the boy was punished for his aggressive behavior.

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In the third film, the third group of children saw that the child was neither rewarded nor punished.

After watching the film, all children were left alone in the room with toys, and those groups of children were observed secretly. In the end, it was observed that those children who had seen the aggressive behavior towards the

toy being rewarded in the film showed the maximum aggression towards the toy. Children who had seen the aggressive behavior being punished showed the minimum aggression.

Therefore, it is clear from this that all children observed the aggression and also imitated it.

It has three factors:-

1) Model learning, 2) Observation or inspection, 3) Imitation.

"In the process of the learning theory, observing a model, imitating its behavior, and internalizing it in one's own behavior to repeat it in the future takes place."

An individual learns in this manner:

i) Paying attention to behavior and understanding it.

ii) Bringing behavior into memory, remembering it.

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iii) Carrying out reinforcement after the behavior, therefore change in behavior for learning takes place only through imitation.

Characteristics of Social Learning Theory:

i) According to this theory, an individual learns only those behaviors on which his attention goes and by which he is influenced.

ii) It emphasizes learning by imitating an individual's behavior.

iii) An individual learns most of the behaviors in his life through the method directed by this theory.

Educational Implications :-

i) The teacher should inspire students for independent observation.

ii) Imitable co-curricular activities should be organized in the school.

iii) The environment of the school should be such that the child can learn by watching.

iv) The teacher should construct unique measures for the welfare of students and assist in children's learning.

* Vygotsky's Socio-Cultural Theory :-

Russian psychologist Vygotsky considered the dialogue between society and cultural relationships crucial in a child's cognitive development.

Vygotsky's socio-cultural theory is also termed a theory of cognitive development.

Like Piaget, Vygotsky (1896-1934) also believed that children construct knowledge, but his view was that development cannot take place in isolation; it takes place along with language development, social development, physical development, and within a socio-cultural context. Piaget's theory is constructivism, whereas Vygotsky's theory is social constructivism.

According to Vygotsky - "Our own development takes place through others."

According to Vygotsky, all mental or intellectual actions first occur externally in society, and through interactions, the child learns the culture (ways of thinking and behaving) of his community. Vygotsky considered various aspects of the social environment, such as family, community, friends, and school, crucial in children's development.

The characteristics of Vygotsky's theory are as follows :-

- i) ZPD (Zone of Proximal Development) holds a major place in the learning process.
- ii) Learning is a universal action that continues throughout life.
- iii) The measures of cultural organizations and psychological facts are responsible for the learning process.

Educational Implications:-

- 1) Self-discipline** - For primary students, Vygotsky believed that maximum socially strong and meaningful actions should be promoted within the child's zone of proximal development, and full attempts should be made to generate self-confidence in them.
- 2) Literacy Activities** - When children formally start going to school, Vygotsky's theory places emphasis on literacy activities.
- 3) Reciprocal Teaching** - This is an educational innovation inspired by Vygotsky's theory in which a teacher and two to four students join together to form a cooperative learning group.
- 4) Collaborative Learning**- It consists of a small group of students, inside which they clear their misunderstandings by providing a sufficient and complete explanation to each other.

UNIT-5 Factors Affecting Learning

* Motivation (Abhiprerana) -

The literal meaning of motivation is—having the willpower to perform any task. Motivation is that internal power which is the product of their stimulus. Through which the child is activated and controlled in the direction of a certain behavior. B. F. Skinner termed motivation as the highway to learning. This is the source of an individual's time and achieving success. Motivation begins with some need and terminates after achieving the goal.

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- According to **Good** - "Motivation is a psychological or internal process that arises in the presence of a certain need."
- According to **Gates and others**- "Motives are those physiological and psychological conditions within the individual that prompt him to perform specific types of actions."

Types of Motivation

i) Intrinsic Motivation

ii) Extrinsic Motivation

1) Intrinsic Motivation: Intrinsic motivation broadly refers to spontaneous interest. When a child learns about a subject, it implies that he has an interest in it. Motivation that generates a sense of self-determination and capacity to adjust with the environment is called intrinsic motivation.

2) Extrinsic Motivation - Extrinsic motivation refers to an incentive given to children in the external environment, shifting their behavior in a certain direction. It has an important contribution to children's learning.

Characteristics:-

1) Motivation is the product of an individual's thoughts and aspirations, 2) Motivation is the main source of fulfilling a need, 3) It is also helpful in character formation, 4) It provides energy to perform any task, 5) This is an excited state of our body, 6) It provides power to our behavior.

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Elements of Motivation :-

1) Positive element- In this motivation, the child performs a task out of his own will.

2) Principles of Motivation :- i) Need for self-actualization, ii) Need for security, iii) Social need - need to remain connected, iv) Physiological need - hunger, thirst, v) Need for esteem - self-esteem, identity.

Educational Significance of Motivation:-

i) For encouragement in learning, ii) For developing discipline, iii) For desired change in behavior, iv) For guidance, v) For social and character qualities, vi) For maximum acquisition of knowledge, vii) Development of interest, viii) Helpful in achieving goals.

*** Attention (Avadhan / Dhyan) :**

We see various kinds of objects in our life, hear about some. We are attracted towards some. Our attention shifts spontaneously. All these behaviors are called attention.

- According to **Ross** -"Attention is the process of getting an object of thought clearly before the mind."
- According to **McDougall** -"Attention is that striving which affects the cognitive process."

Characteristics of Attention:

i) Mental activation, ii) Fickleness / dynamism, iii) Narrowness, iv) Selectivity, v) Readiness, vii) Heuristic/exploratory, viii) Analytical, ix) Synthetic.

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Focusing Attention:

i) Peaceful environment, ii) Complete preparation of the lesson, iii) Variation in the subject, iv) Use of audio-visual aids, v) Appropriate behavior towards children, vi) Giving a place to the interests of the child, vii) Knowledge of the child's tendencies.

Factors Affecting Attention:-

1) External determinants, 2) Internal determinants.

1. External determinants: External determinants consist of the following types: (i) Movement, (ii) Intensity, (iii) Duration, (iv) Size, (v) Novelty/Contrast, (vi) Repetition, (vii) Change, (viii) Nature.

2. Internal determinants: Internal determinants consist of the following types: (i) Interest, (ii) Emotion, (iii) Need, (iv) Goal, (v) Habit, (vi) Past, (vii) Experience, (viii) Training, (ix) Instincts.

Measures to Focus Children's Attention in Educational Work:-

- 1) The teacher should infuse so much confidence and hope in students so that they can pay attention towards the subject matter being taught out of their own will.
- 2) To attract students' attention towards themselves, teachers should take the help of an appropriate teaching method.

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- 3) Teachers should keep the environment of the school and class such that the student does not experience any kind of fear or obligation.
- 4) To attract students towards themselves in class, teachers should also use some toys and hand-made pictures.
- 5) To attract students' attention, teachers can use external determinants of attention.

* Memory (Smriti) :-

Recalling any learned subject matter or action when required is called memory.

In psychological terms, memory is a mental process that is definitely found in some amount or another in every living being.

- According to **Woodworth** - "Memory is the direct use of what has been learned."

"Memory is a mixed process in which the imitation of the state of mind occurs."

Concept of Memory:-

- i) Process of learning, ii) Retaining the action of learning, iii) Recalling the learned action, recognizing the recalled action, iv) Recognition - providing completeness to memory.

Neurophysiological Psychology (Neurodaihiik Vigyan):-

Among the various branches of psychology, neurophysiological psychology is an important branch. Through this, assistance is obtained not only in understanding the complexities of human behavior theoretically, but the knowledge obtained from neurophysiological psychology is also successfully used in removing various types of behavioral abnormalities. It is a science that studies on the basis of neurophysiology.

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Neurophysiological psychology is also known by the names of biological psychology and psychobiology. Its origin took place in educational areas. Neurophysiological psychology is also interested in studying the impacts of hormones on behavior.

The basis of human behavior is biological structure. Biological structure is both internal and external. Under this, the functional systems of anatomy are studied. In physiology, we study nerves, blood pressure, digestive system, etc.

It is written in the Encyclopedia Britannica, "Physiological psychology is the science that expresses the relationship of behavior and mental processes with the physical actions of various organs of the body."

Parts of the Human Brain :-

The brain has three parts –

1. Forebrain, 2. Midbrain, 3. Hindbrain.

1. Forebrain - It consists of two parts. The cerebrum is made up of two hemispheres. It also consists of four parts. This is the center of intelligence, consciousness, power, and memory power. From this part of the diencephalon emerge the pineal body and pituitary gland.

2. Midbrain - It is made up of two parts: the cerebral peduncle and corpora quadrigemina. It exercises control over vision and auditory power.

3. Hindbrain - This is the hindmost part of the brain, which is composed of the cerebellum and medulla oblongata; it maintains the balance of the body.

Educational Implications :-

- 1) Biological processes also affect the learning capacity of children.
- 2) All children have different preferences for learning. Teachers should perform teaching work according to these preferences.
- 3) The more flexibility children possess towards various concepts and ideas, the more quickly they learn new concepts.